MANCHESTER 1824
The University of Manchester

EART11100 Understanding the Earth

Semester 1: The Earth's Evolution

How to Build a Planet: Evolution of the Physical Earth System

How to Grow a Planet: Evolution of Life on Earth

Workshop: Geological Time and Big Ideas

Semester 2: The Present Earth

A Dynamic Planet

A Sustainable Planet

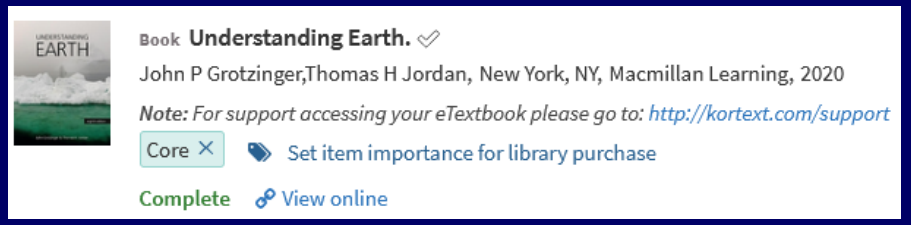
Lectures

Background reference frame for all pathways from Year 2

We will deliver more content than will be assessed

- Know by heart the principal divisions of the stratigraphic column
- Have an outline familiarity with key events in the history of the Earth and Universe
- Use the weekly study guide to organise your revision notes

Live Class: Monday, 10am-12pm



- Attendance is compulsory (SEAtS)
- Pre-lecture reading (textbook)
- Lecture handouts will be on Blackboard before the class
- Podcasts are available after the lecture

Semester 1 Assessment (50% of the course unit): Build and Grow

- Formative Test in Week 6
- Summative Exam in January

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How to Build a Planet

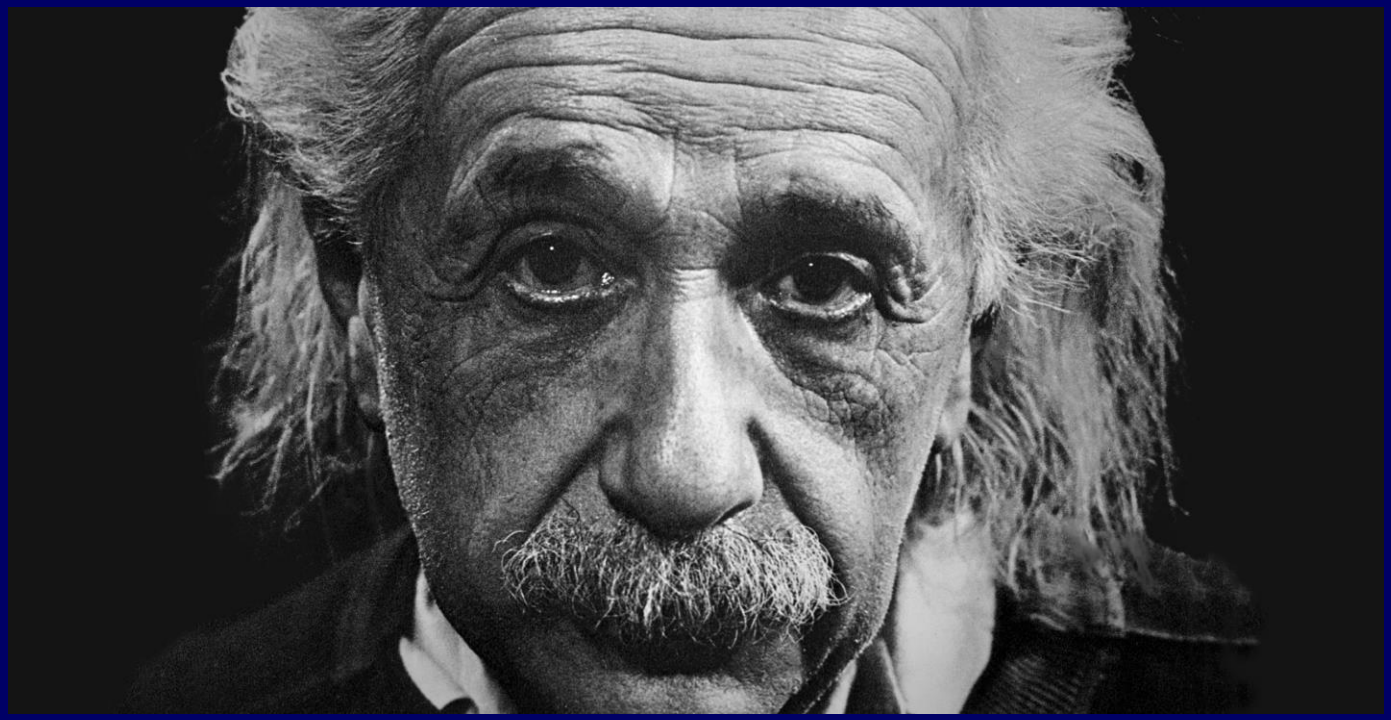
Before the Earth:

The Formation of the Universe

13797-13650 Ma

In the beginning……

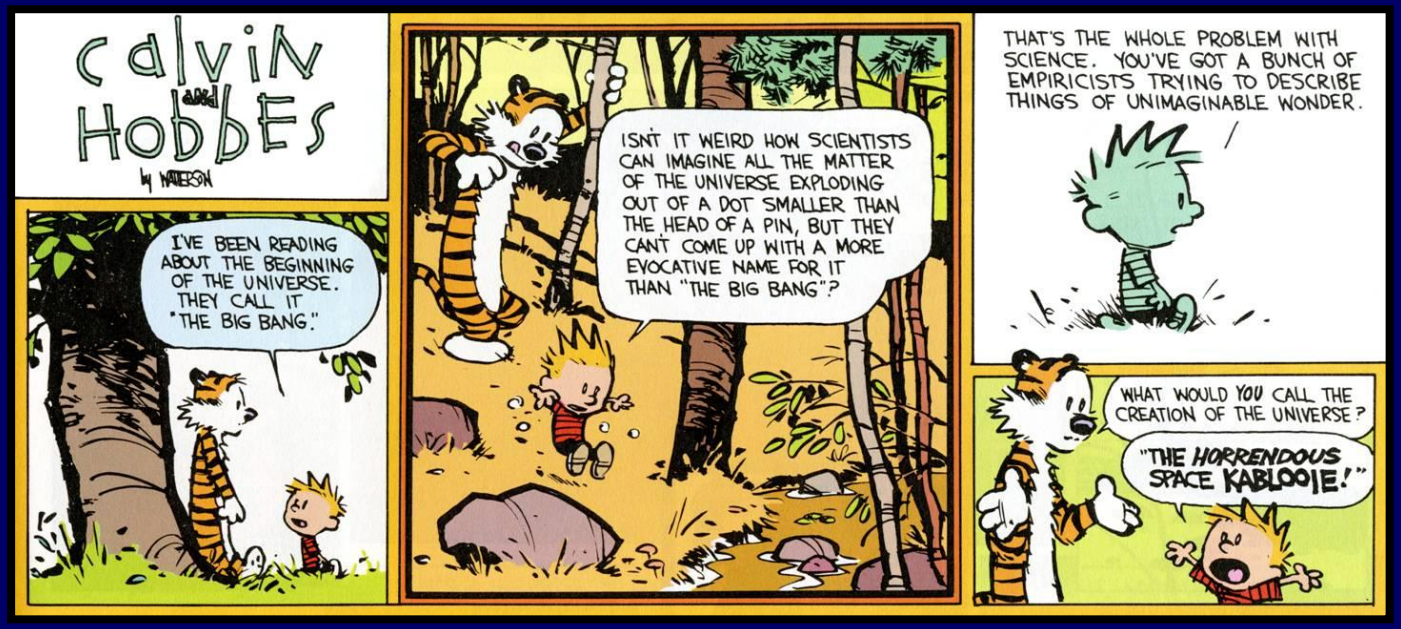
The Standard Model of Big Bang cosmology (the Λ-CDM model) includes a cosmological constant, denoted by lambda (Λ), associated with dark energy, cold dark matter, denoted by CDM and ordinary matter.



Albert Einstein (1879-1955)

The Big Bang: t = 0 s

- ‘before the Big Bang’ all matter, energy and space in the Universe was compacted into an inconceivably small, hot and dense point (a *singularity*)
- erupted in a huge cosmic ‘explosion’
- space, time, energy and matter began expanding everywhere at once
- current best estimate of this time is 13797 Ma (= age of Universe)



The hot, dense earliest moments of the Universe are described as ‘The Big Bang’

What's in a name?

The Big Bang addresses the *evolution*, not the *creation*, of the Universe

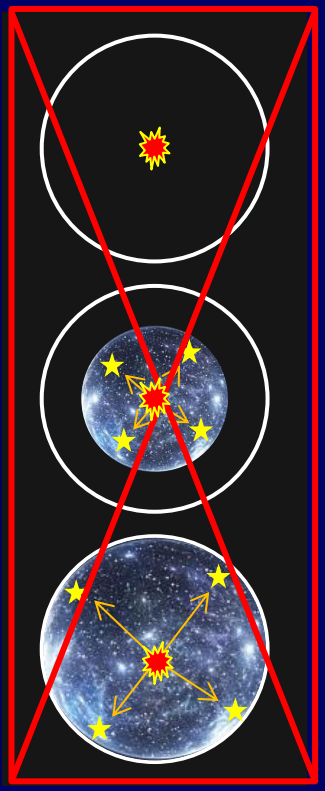


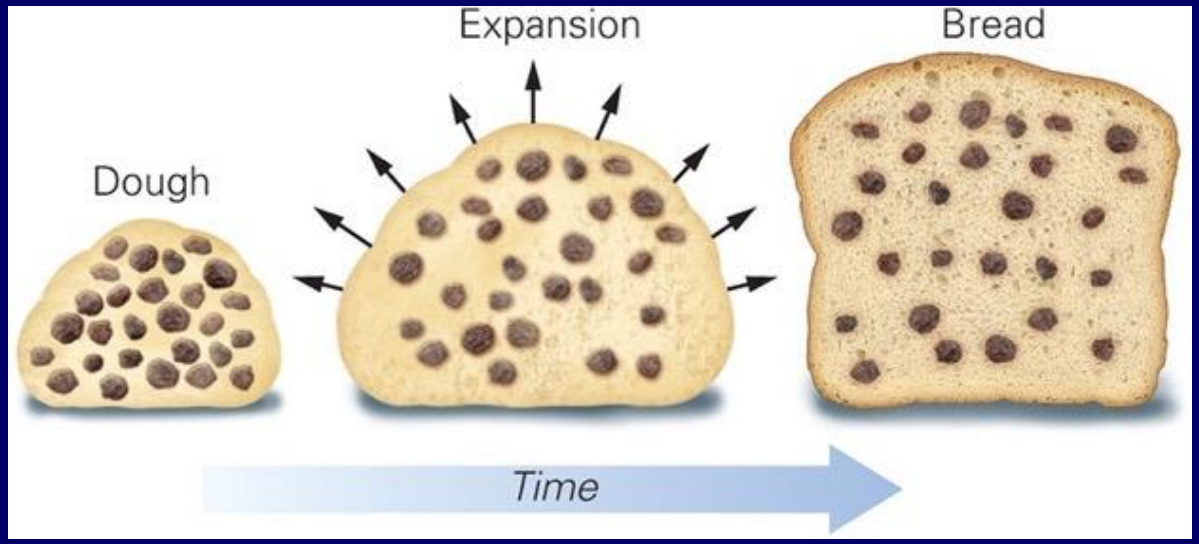
Fred Hoyle (1915-2001) British astronomer who coined the phrase 'The Big Bang'

Sir Fred Hoyle formulated the theory of stellar nucleosynthesis but rejected the Big Bang theory in favour of a steady state universe

Visualising The Big Bang

- neither ‘Big’
- nor a ‘Bang’
- stretching *with* space, not expanding *into* pre-existing space
- no origin of expansion





Raisin dough analogy for the expansion of space

But what is the evidence for an expanding Universe?

Red star at night… Hubble's delight!





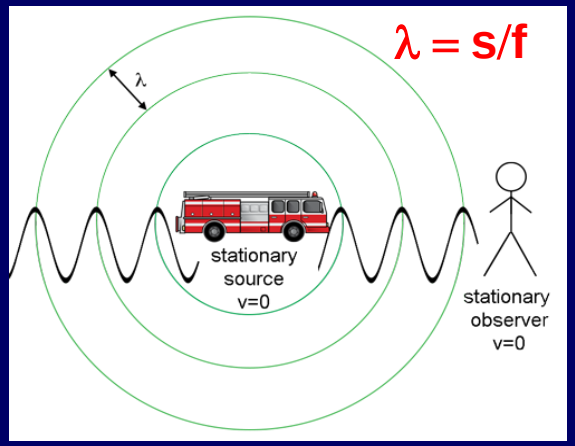


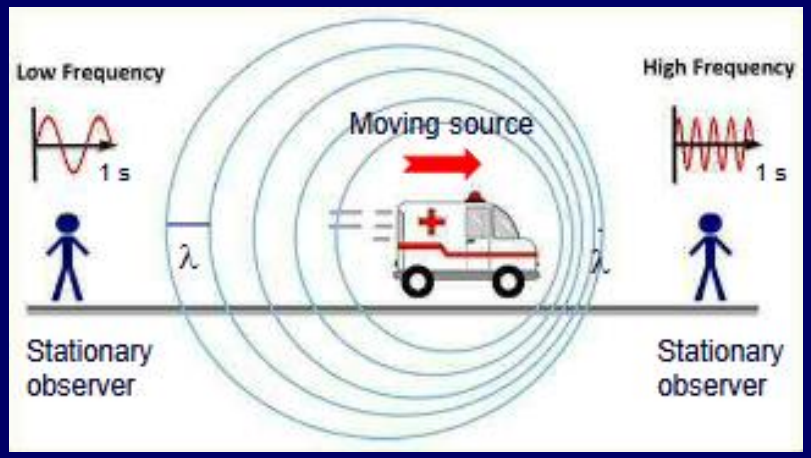
Edwin Hubble
(1889-1953)
Astronomer

Light from distant galaxies are slightly red in colour due to 'red shift'

Red shift is commonly explained as a ‘Doppler effect’

Movement of the source alters the wavelength (λ) and the received frequency (f) and pitch of sound by a stationary observer, even though the source frequency and wave velocity (s) are unchanged

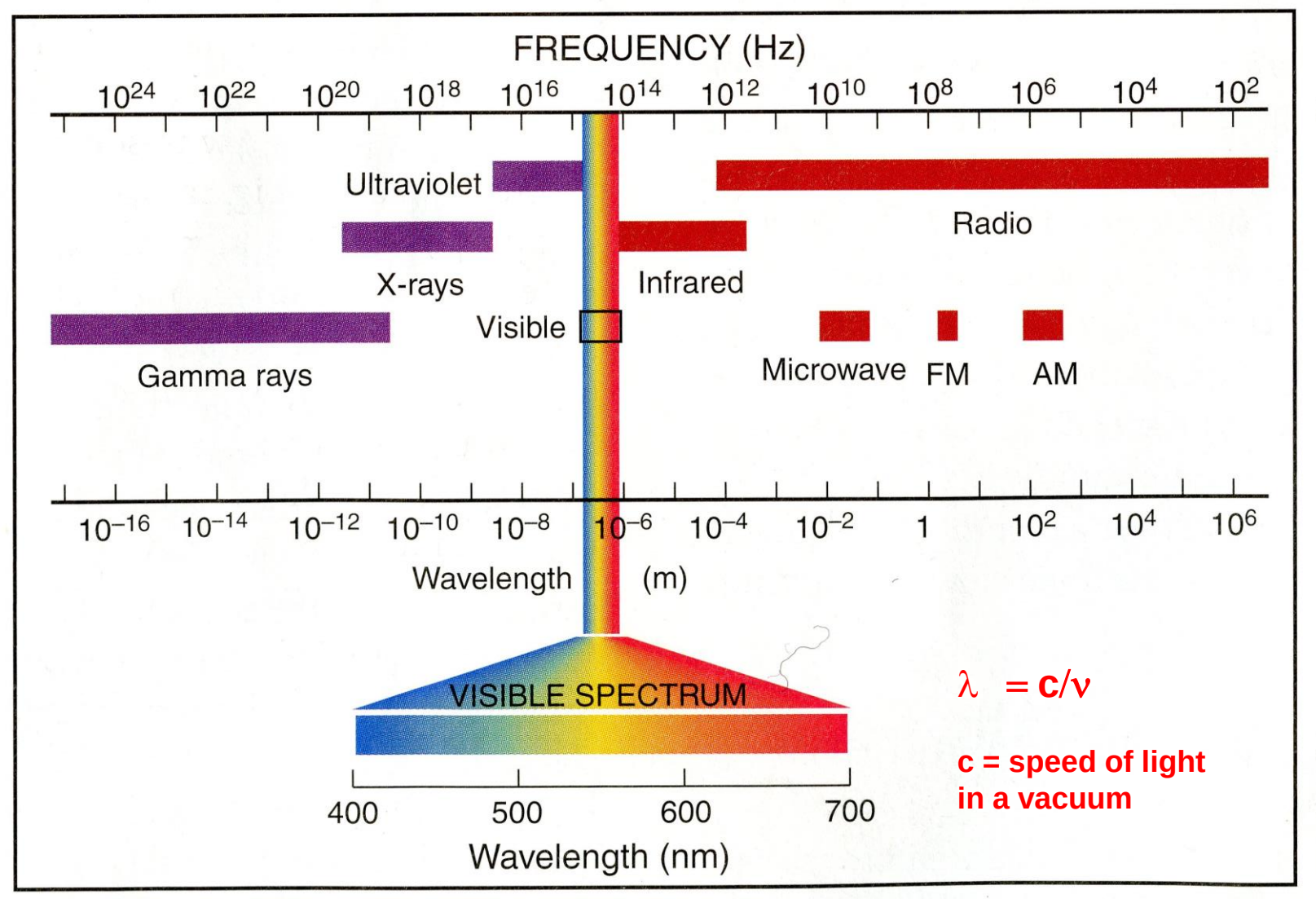






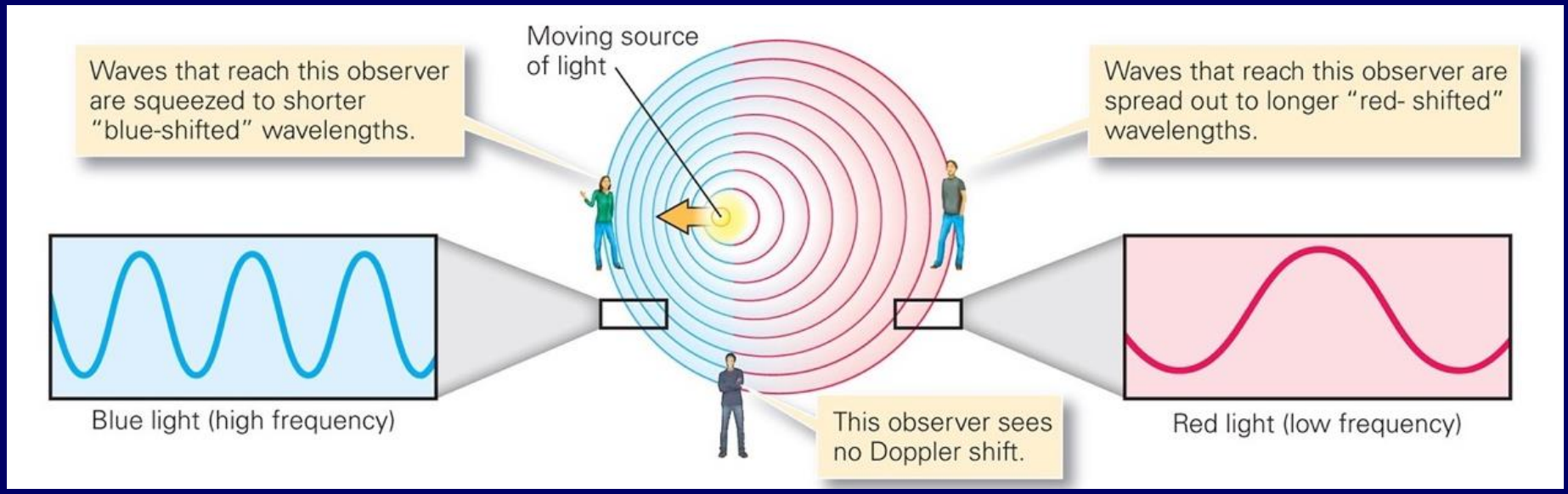
Like sound, light can be thought of of consisting of waves

Frequency and colour of visible light



The electromagnetic spectrum

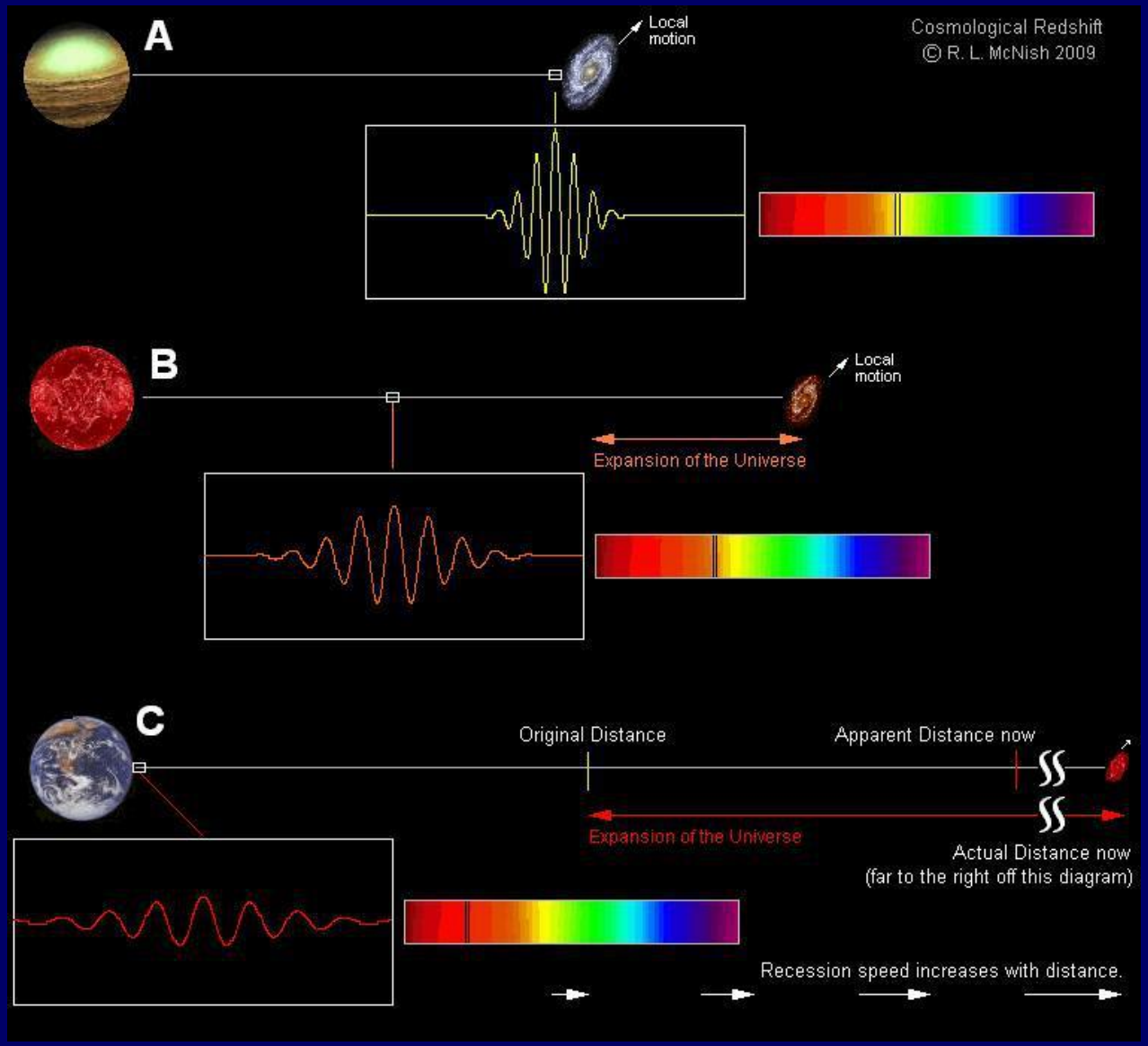
Doppler redshift

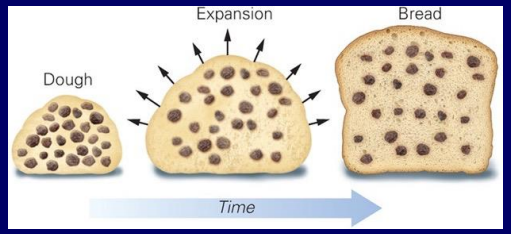


Doppler redshift due to the light source moving relative to the observer

BUT
Astronomical redshifts are 'expansion redshifts' due to the expansion of space itself

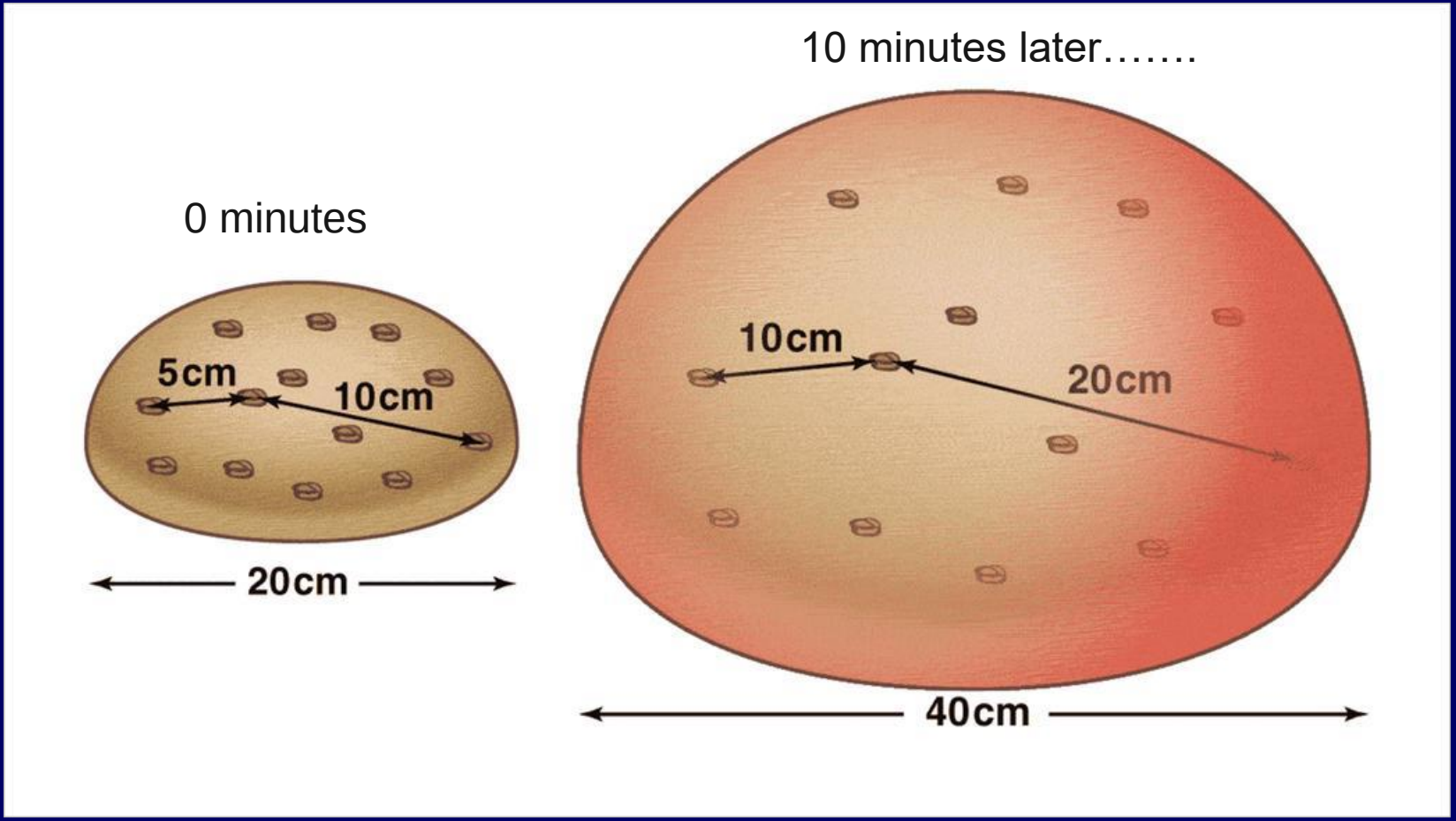
Redshift due to an expanding universe





Raisin dough analogy for the expansion of space

Hubble's Law



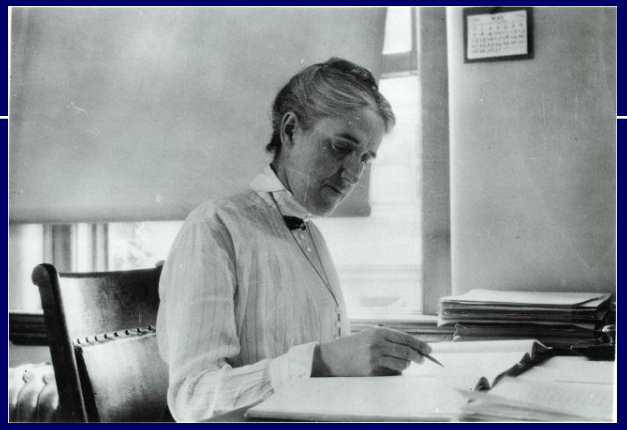
Objects in space move apart at a rate that increases with the distance between them

Hubble's Law

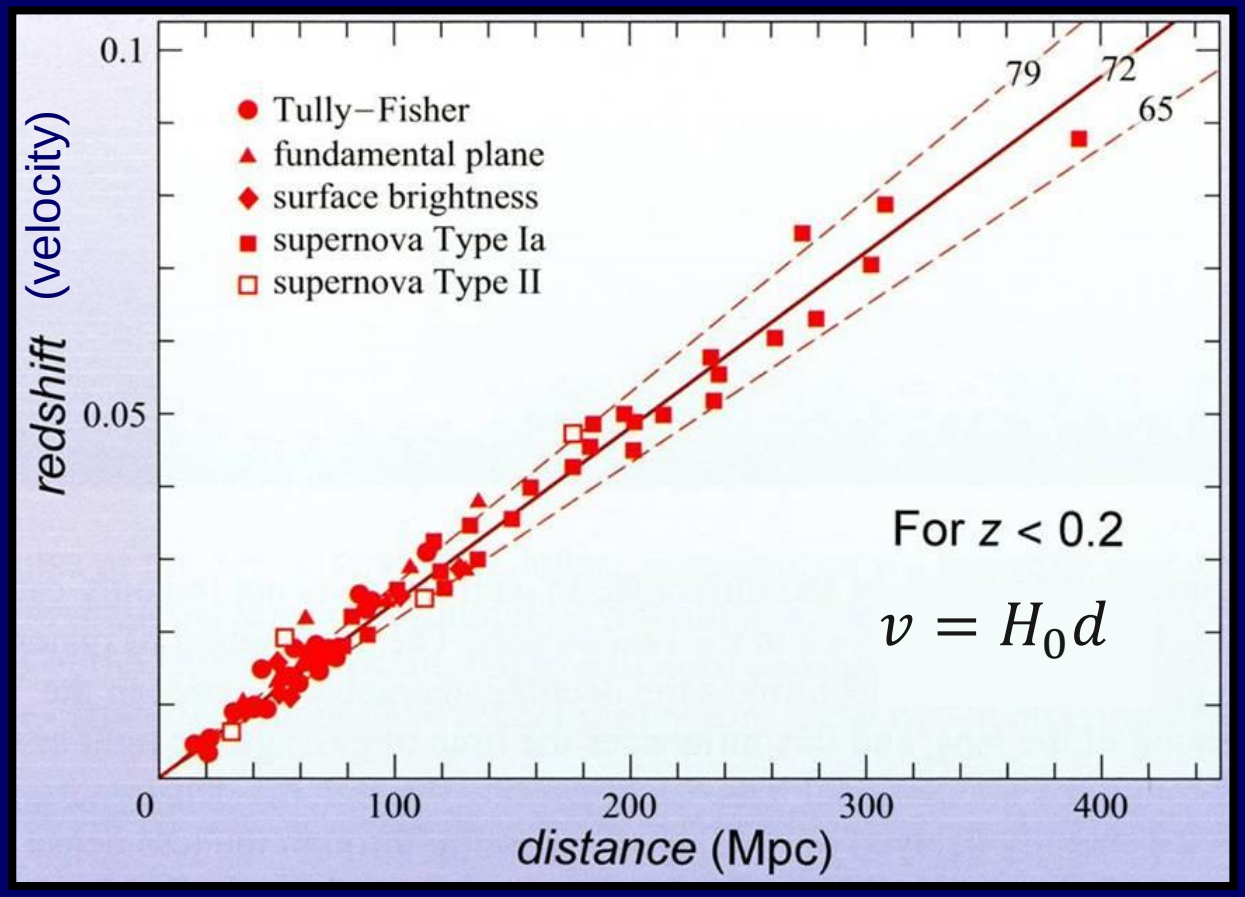
$$H_0 \approx \frac{70\,km/s}{Mpc} \approx \frac{20\,km/s}{Mly}$$

Mpc = million parsecs

Mly = million light years



Henrietta Swan Leavitt
(1868-1921)
Astronomer



Distances estimates based on brightness (absolute and apparent)

- Cepheid Stars
- Spiral Galaxies
- Elliptical Galaxies
- Supernovae

H_0 = Hubble constant

d = distance

v = apparent speed

Hubble's constant and the age of the Universe

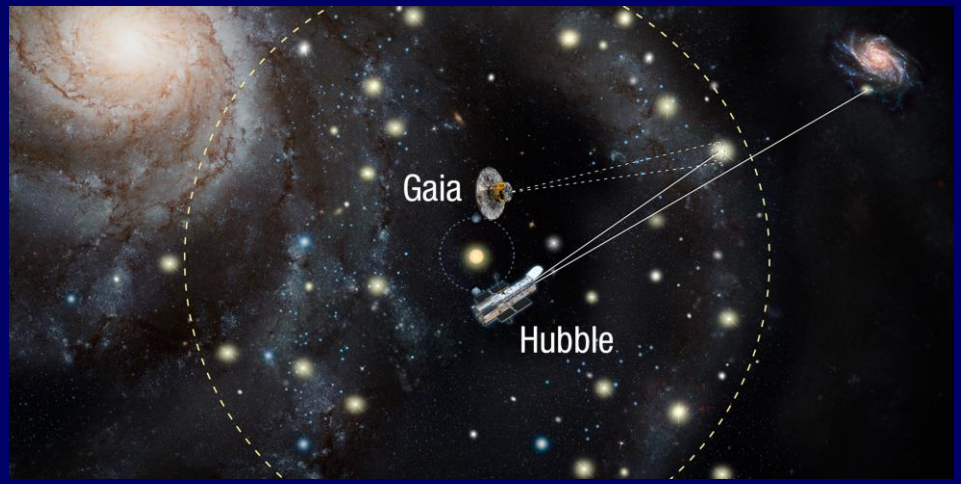


Image: NASA, ESA, and A. Feild (STScI) NEWS RELEASE: 2018-34 >

NASA and ESA Gaia and Hubble

- H_0 of nearby universe
- based on measurement of Cepheid variable stars
- 73.5 $kms^{-1}Mpc^{-1}$

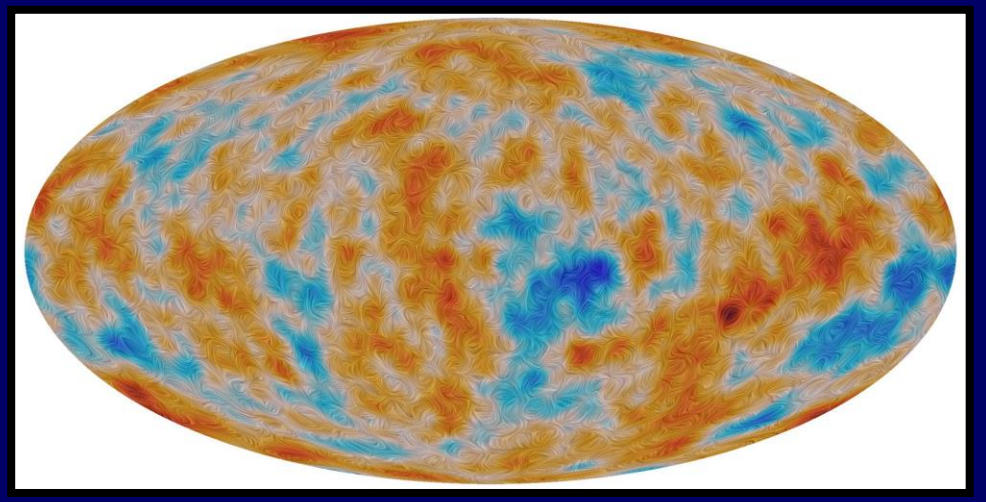
$$t = \frac{d}{v};\ v = H_0 d$$

$$t = \frac{d}{H_0 d} = \frac{1}{H_0} = \left(\frac{71}{3.09 \times 10^{19}}\right) \times \frac{1}{s}$$

= 4.3521127 x 10^{17} s = 13.8 Ga

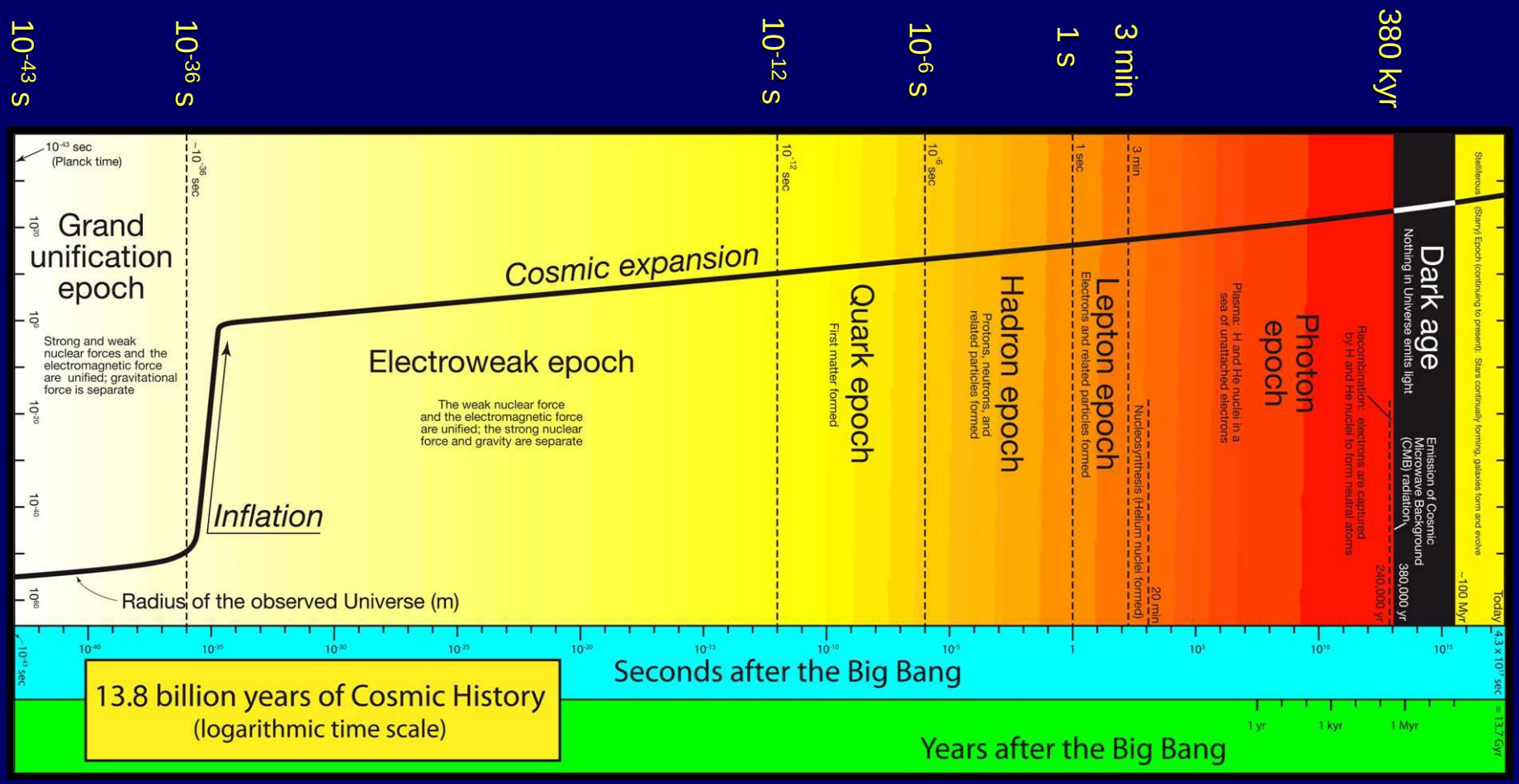
ESA Planck mission

- H_0 of distant, primal universe as it appeared 360,000 after the big bang
- based on model using Cosmic Microwave Radiation
- 67.0 $kms^{-1}Mpc^{-1}$



Could there be new physics underlying the foundations of the universe?

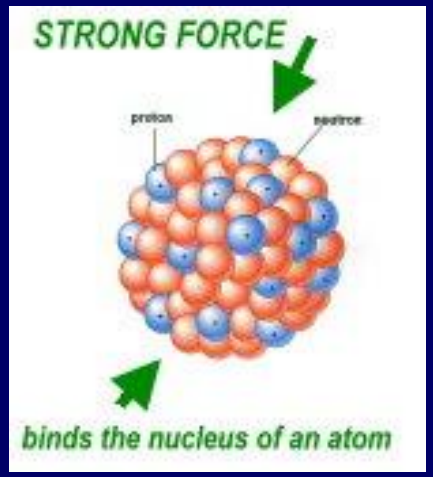
Following the Big Bang……



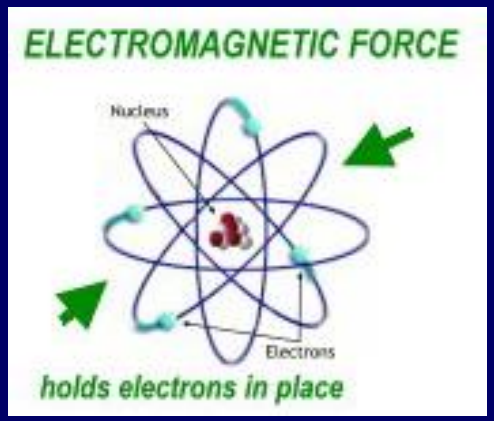
Expansion and Cooling

Plank Epoch: The four forces are unified: 0 s $< t \leq 10^{-43}$ s

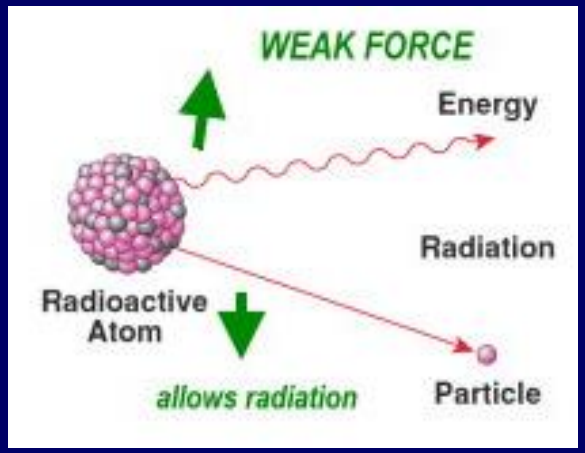
Era of Quantum Gravity or 'Theory of Everything'



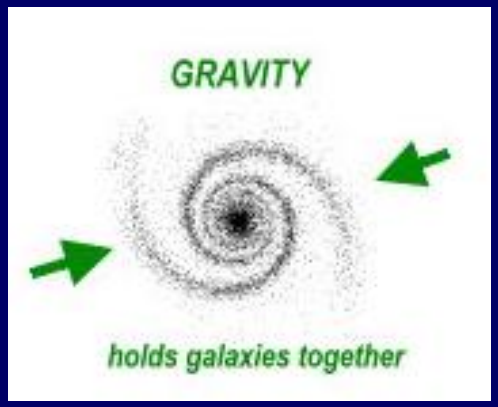
Atomic Scale



Atoms, molecules, electricity, magnetic fields

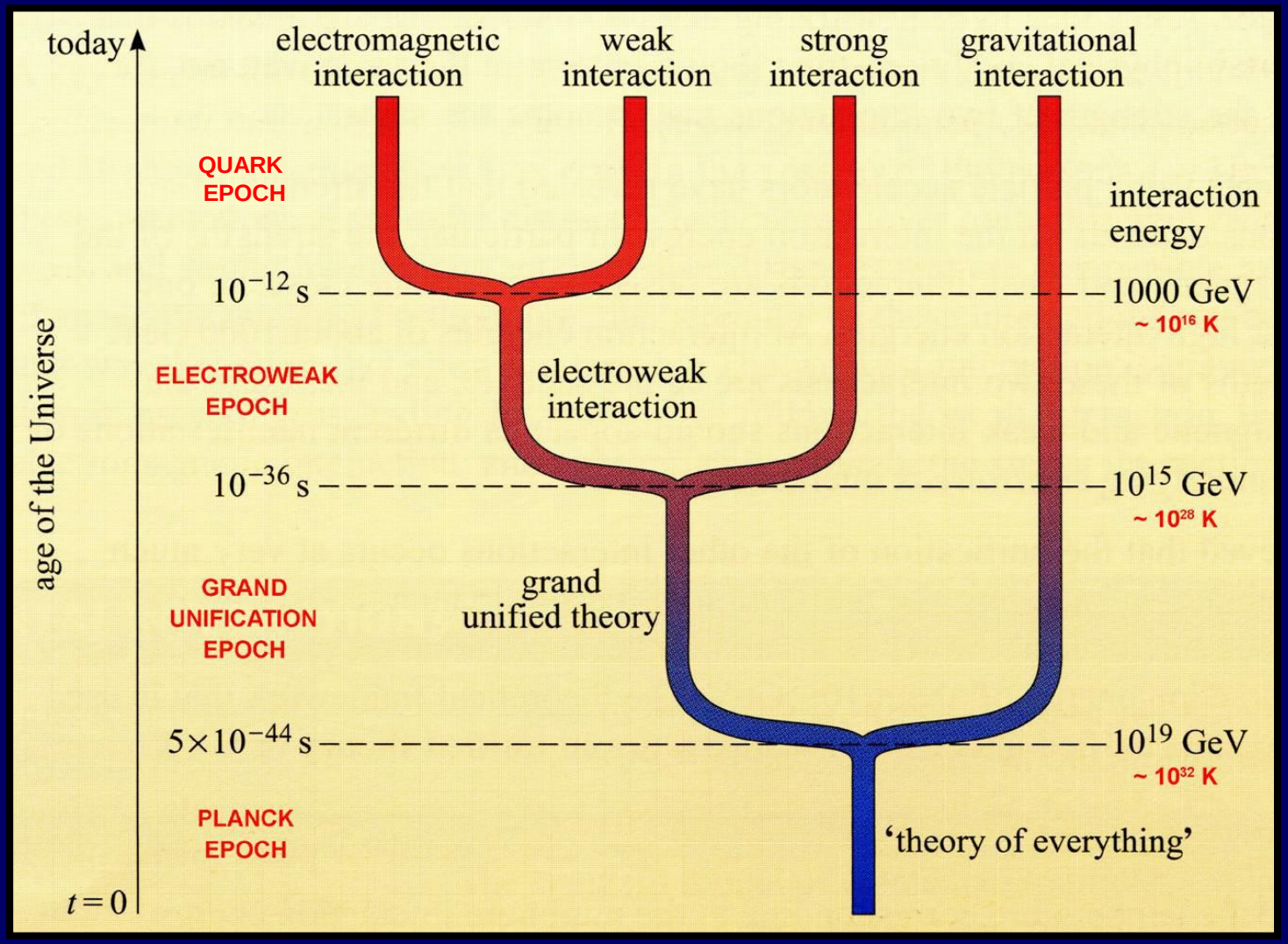


Atomic Scale



Motions of planets, stars, galaxies

Theory and observation

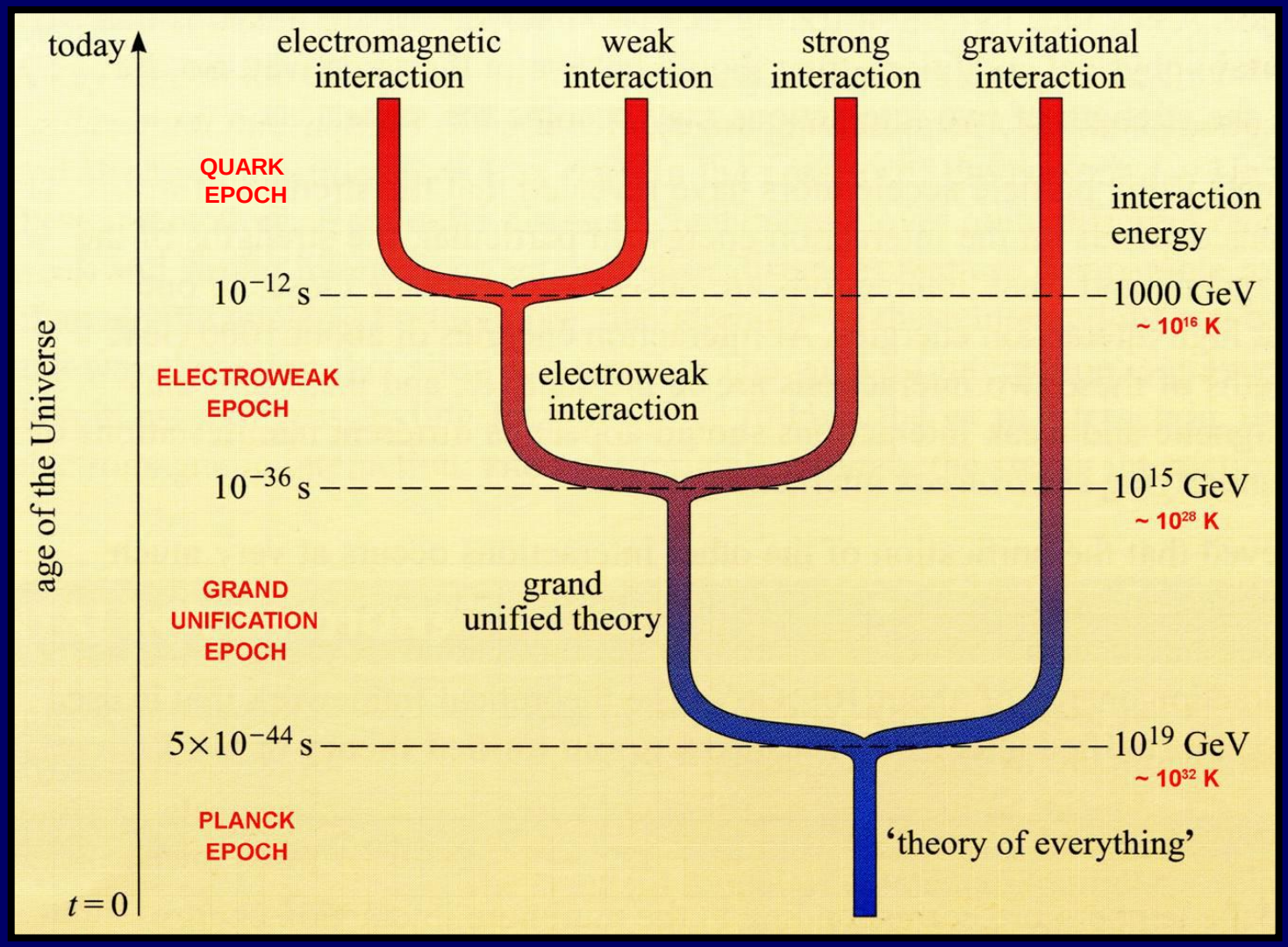


Secure theory

6000 GeV (LHC)

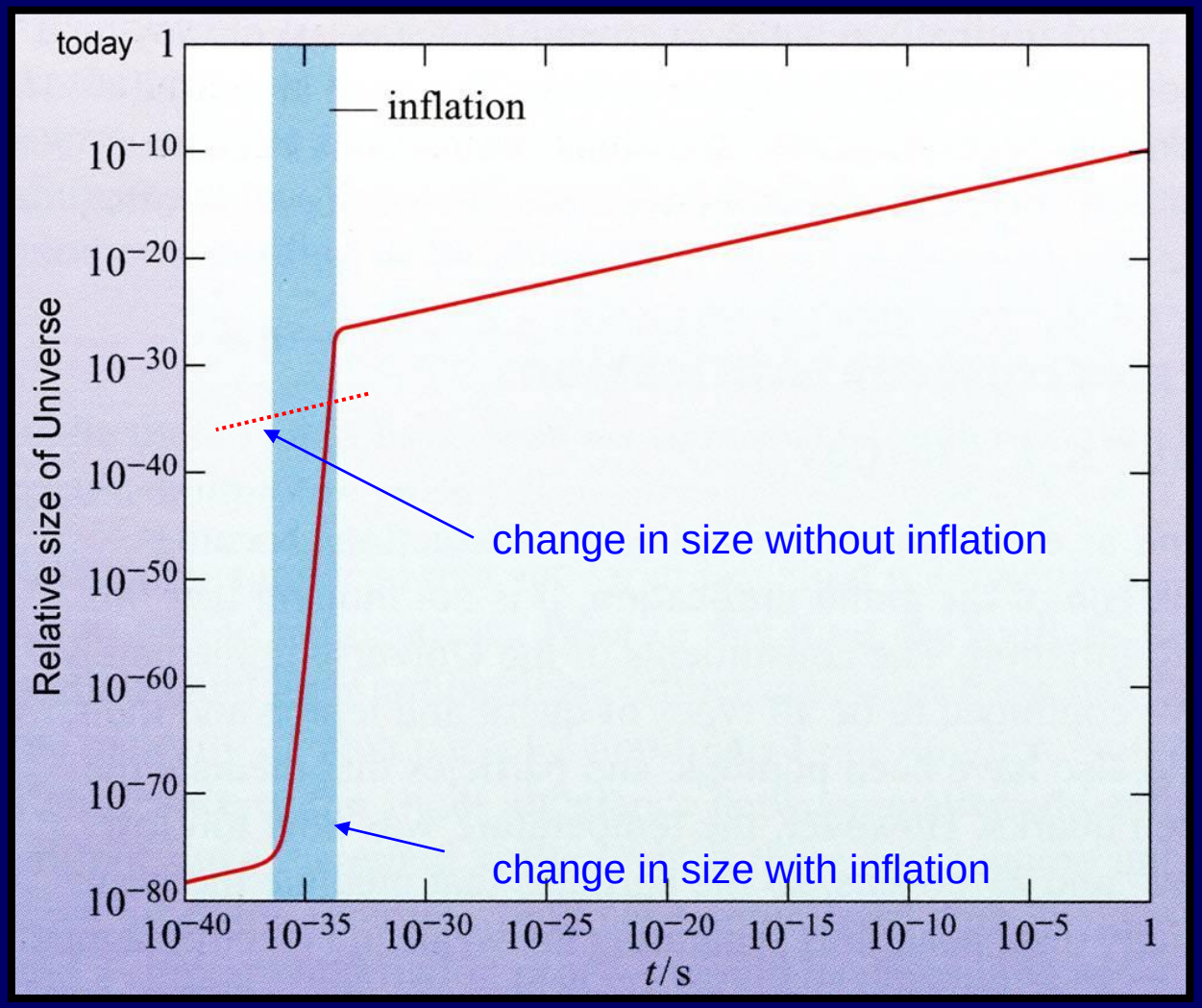
Speculative theory

The gravitational force separates: 10^{-43} s < t ≤ 10^{-36} s



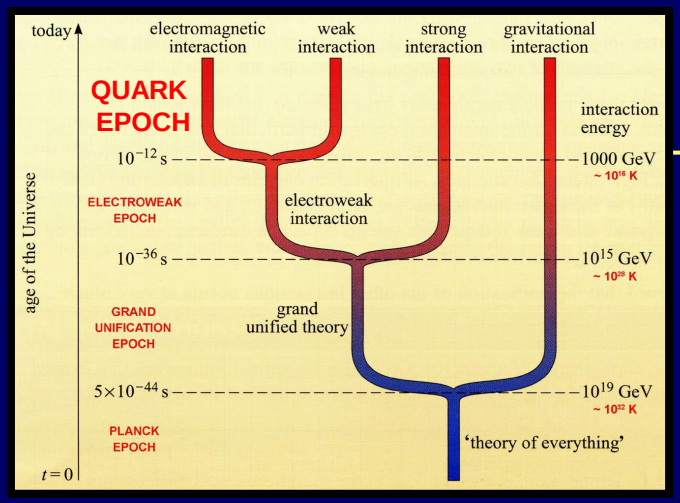
At 10^{-43} s the Universe spanned a region of ≤ 10^{-35} m and was at a temperature of 10^{32} K

EW Epoch: Inflation of space-time: 10^{-36} s < t ≤ 10^{-32} s



- Inflation of space-time occurred faster than the speed of light!
- The Universe increased to grapefruit size (a factor of at least 10^{27} increase in diameter) within 10^{-32} s

Quark Epoch: The four forces are distinct: 10^{-12} s < t ≤ 10^{-6} s



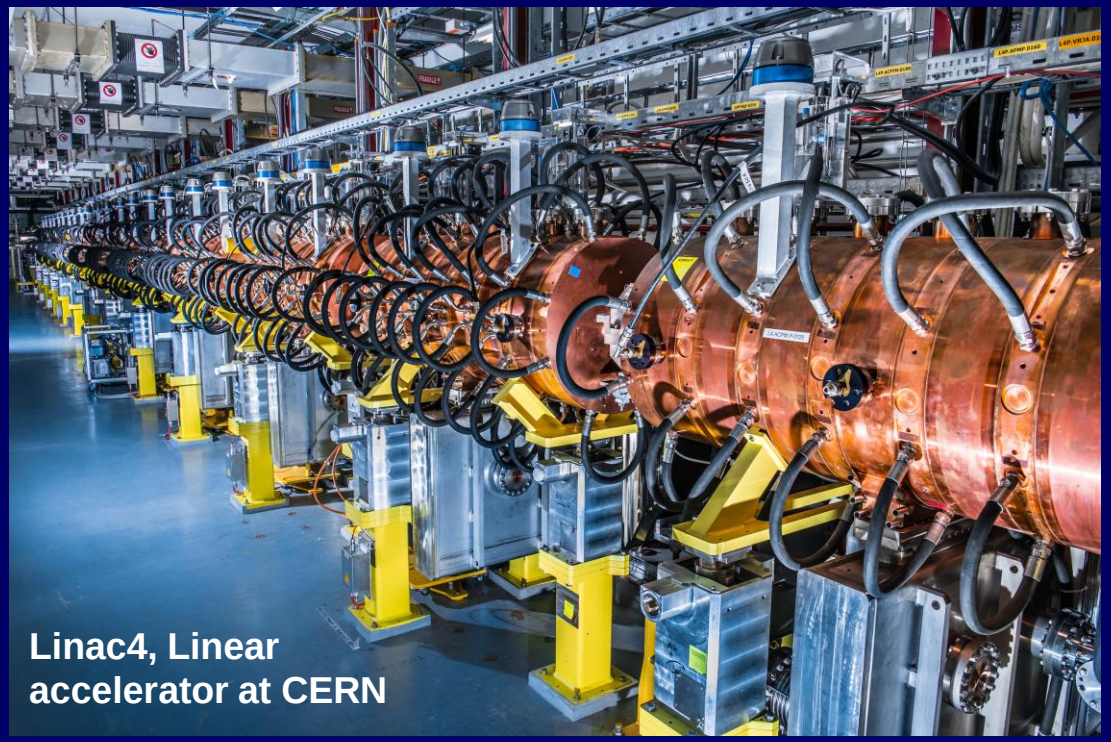
Secure theory

Speculative theory

By 10^{-12} s, the Universe had cooled to a temperature of 10^{15} K



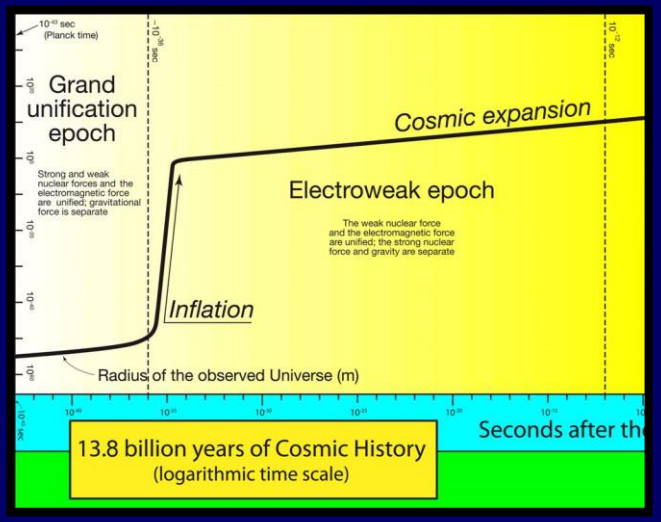
Quarks: an elementary particle and major constituent of matter



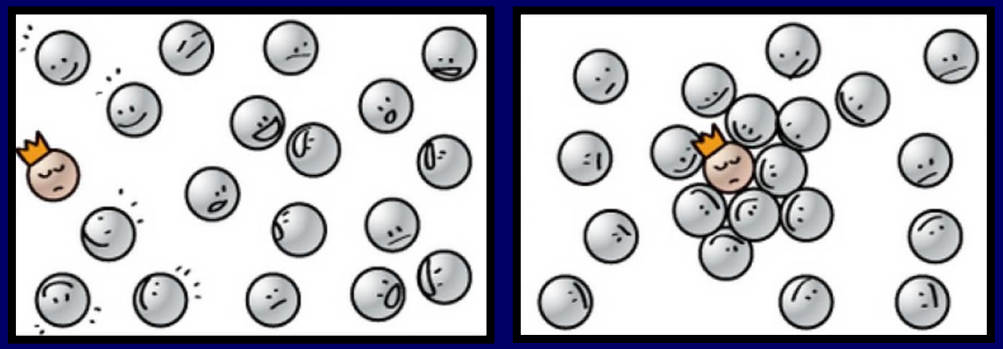
Linac4, Linear accelerator at CERN

The appearance of mass: $t \sim 10^{-12}$ s

- The Universe is permeated with an energy field (Higgs Field)
- If it did not exist no particles would have mass and would move at light speed
- Just after the Big Bang the Higgs field was zero
- As the universe cooled, the field grew spontaneously
- As other particles pass through the field, their interactions with Higgs bosons confers mass on them



- The visible manifestation of energy fluctuations in the Higgs field is the Higgs Boson

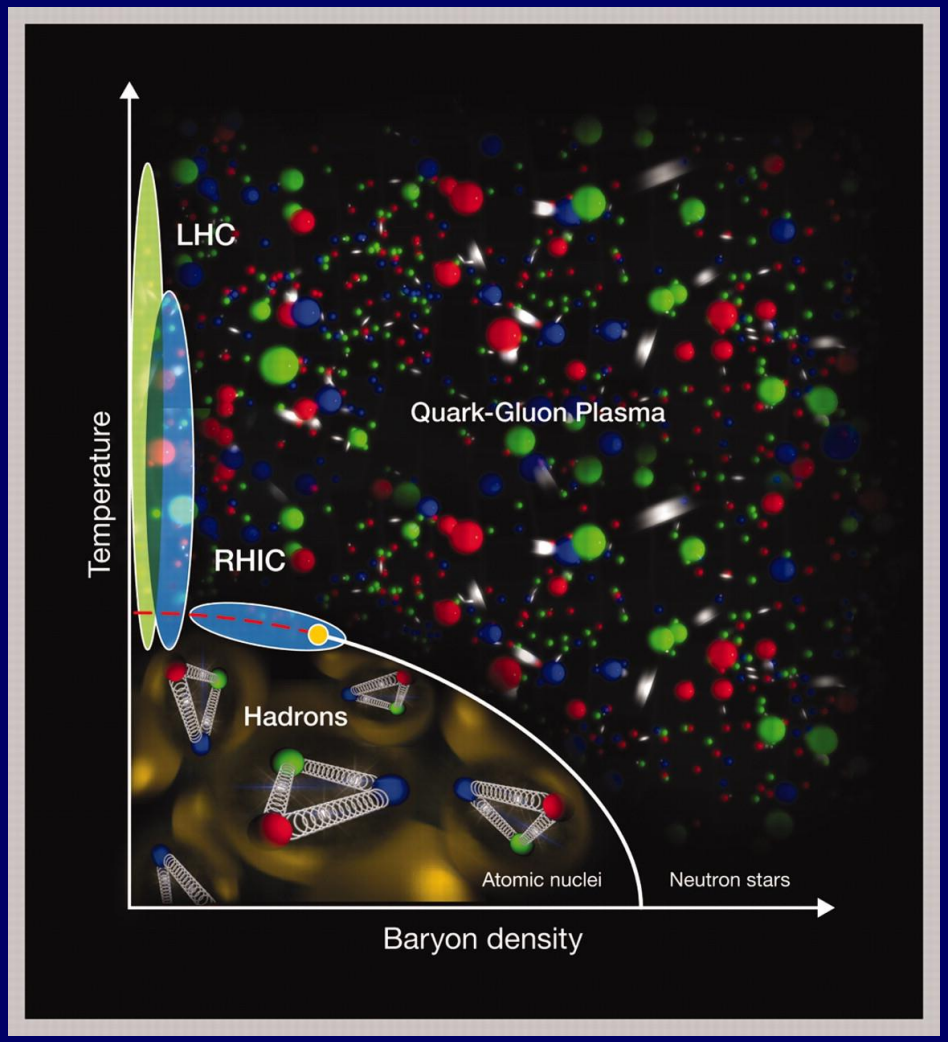


mass is a resistance to force (or via $F = ma$) to acceleration

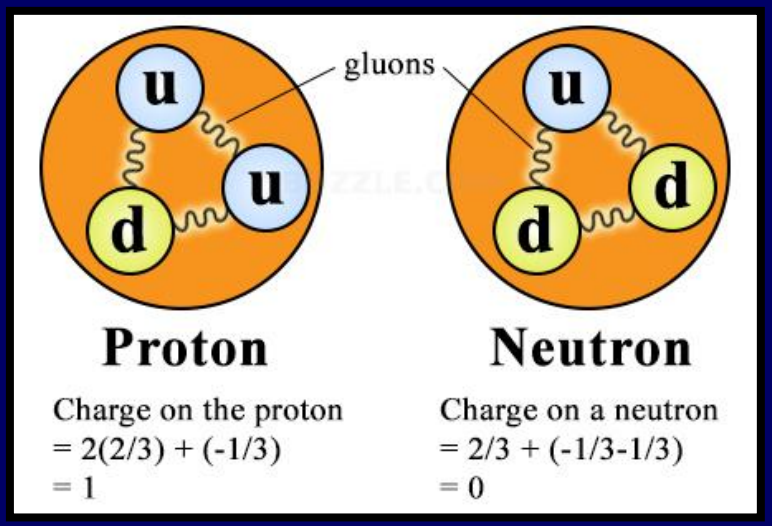
A cocktail party analogy

If the guest-filled room is the Higgs field, clumps of guests within it are Higgs bosons. The passage of another particle (celebrity or otherwise) across the room is slowed down (experiences resistance) as it has to negotiate its way around and through the clumps of other guests- it has acquired mass

Hadron Epoch: The first normal matter forms: 10^{-6} s < t ≤ 1 s



Phase diagram in the temperature-baryon density plane



The quark structure of protons and neutrons

By 10^{-6} s, the Universe had cooled to a temperature of 3×10^{12} K

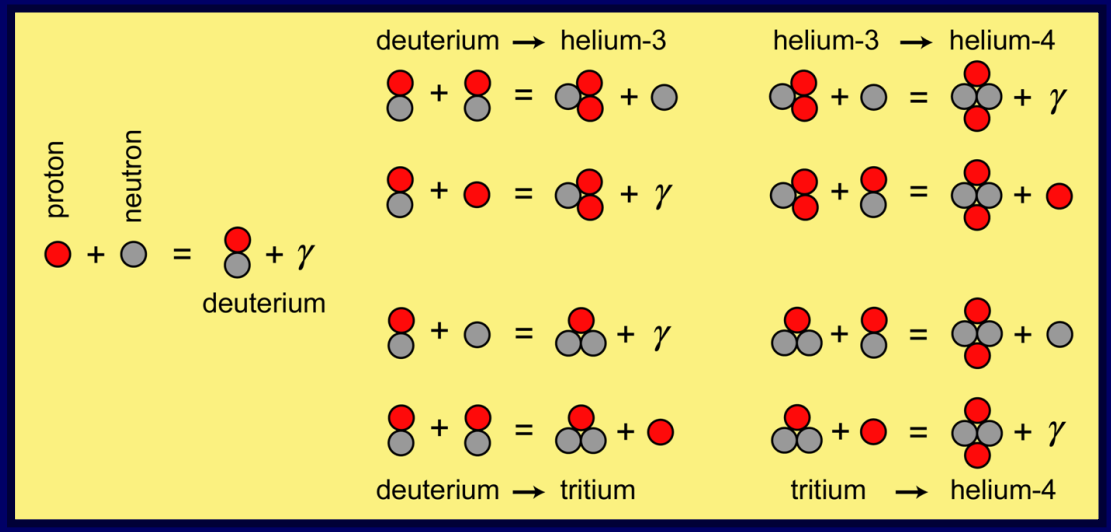
- the first normal matter formed: quarks combined to form hadrons (including protons and neutrons)

Photon Epoch: Primordial nucleosynthesis: 3 min < t ≤ 20 min

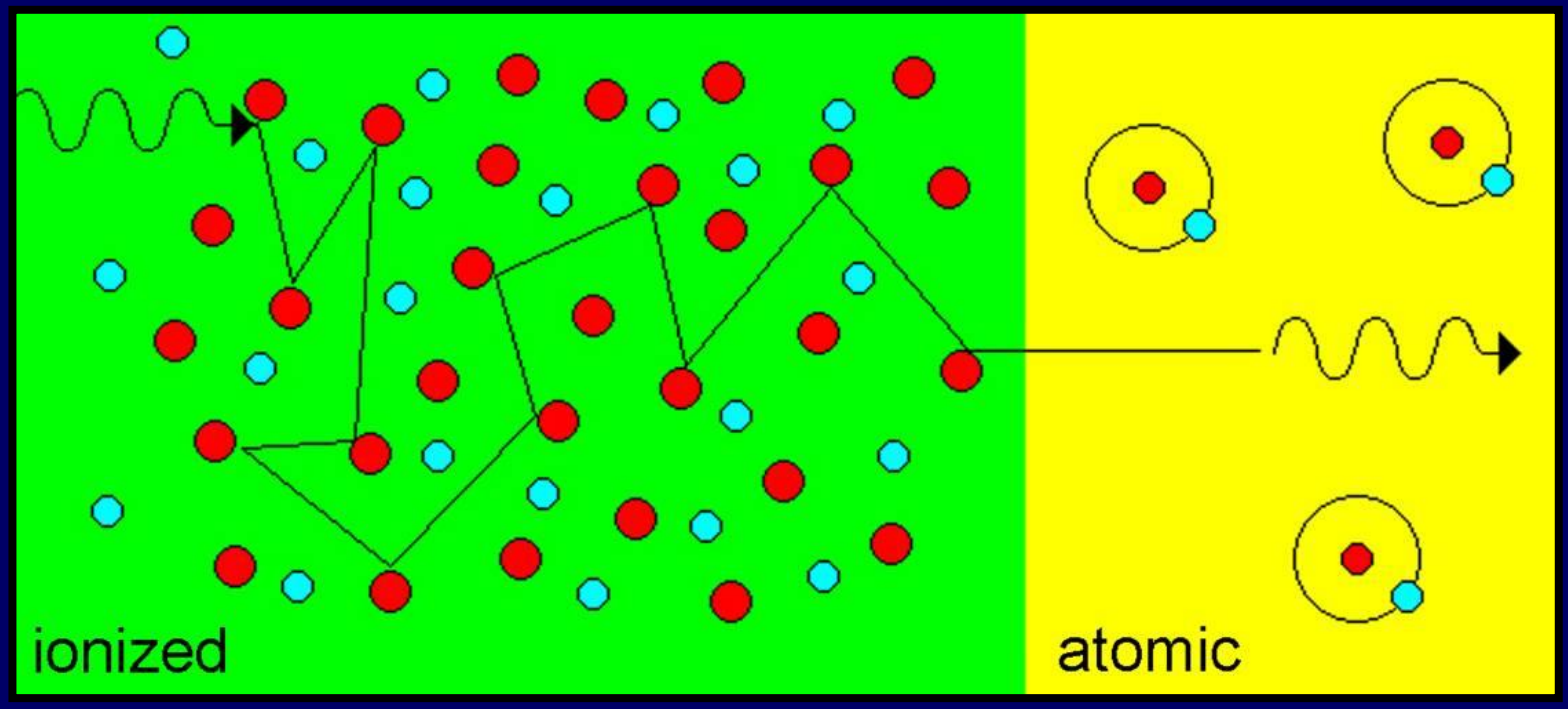
- By 3 min, the Universe had cooled to a temperature of 10^9 K
  - protons and neutrons combine through nuclear fusion to form simple elements of hydrogen, helium and lithium
- By ~ 20 min the temperature and density of the Universe had fallen to values where nuclear fusion could not continue
  - the elemental abundances in the Universe were then set at ~ 75% hydrogen and ~25% helium with trace lithium, abundances that have been only slightly changed through subsequent stellar nucleosynthesis

The dominant fusion reactions that led to the production of helium-4

Primordial nucleosynthesis effectively stopped at helium-4



A transparent universe emerges: 3 min < t ≤ 380 kyr



Plasma of free (ionized) nuclei and electrons under intense photon bombardment (plus dark matter)

Before ‘recombination’ the photons were trapped and the Universe was opaque

Formation of neutral atoms

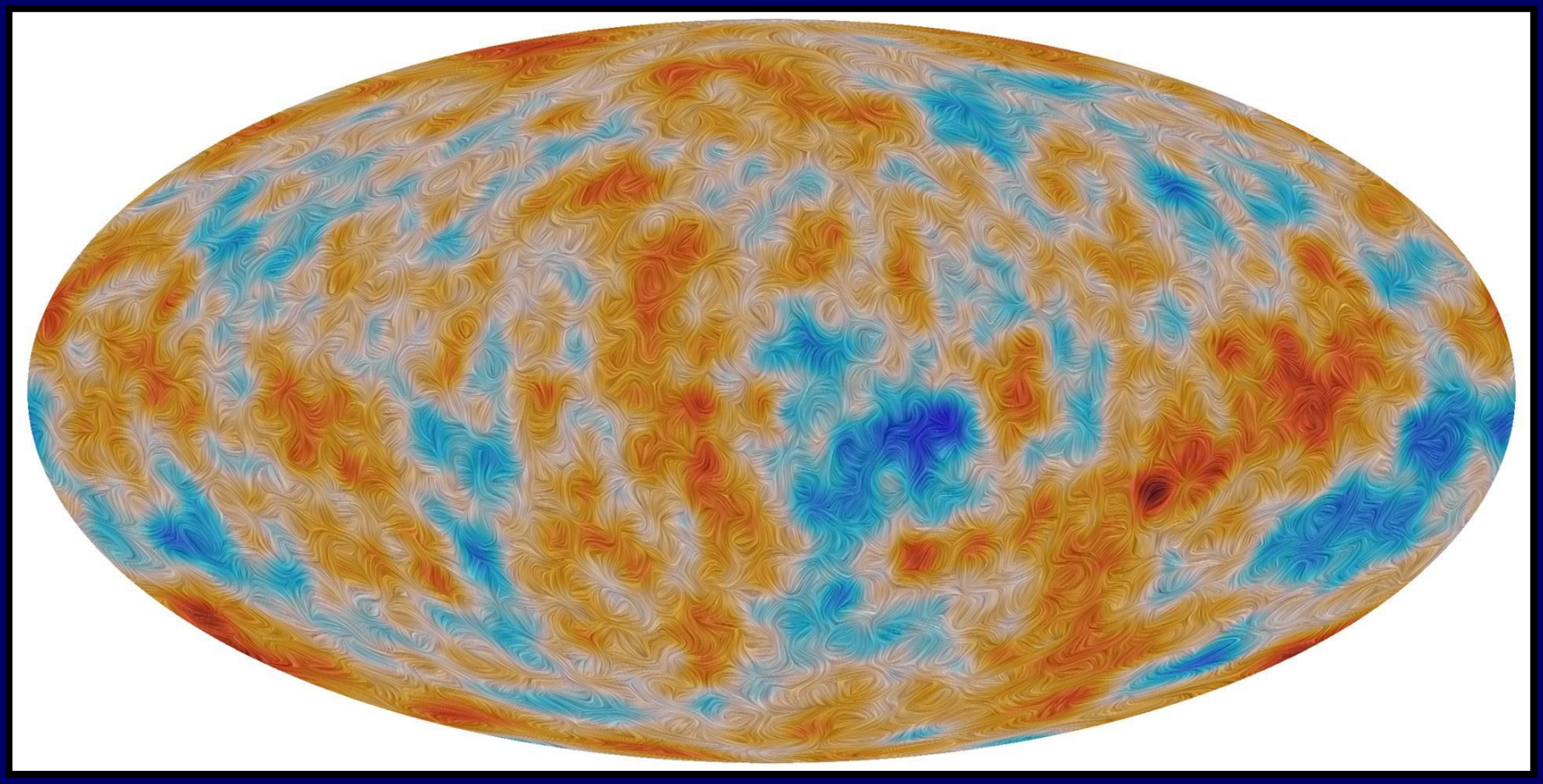
Capture of electrons forming neutral atoms (recombination)

Once recombination had begun, the Universe became transparent

Photons emerging after recombination formed the cosmic microwave background

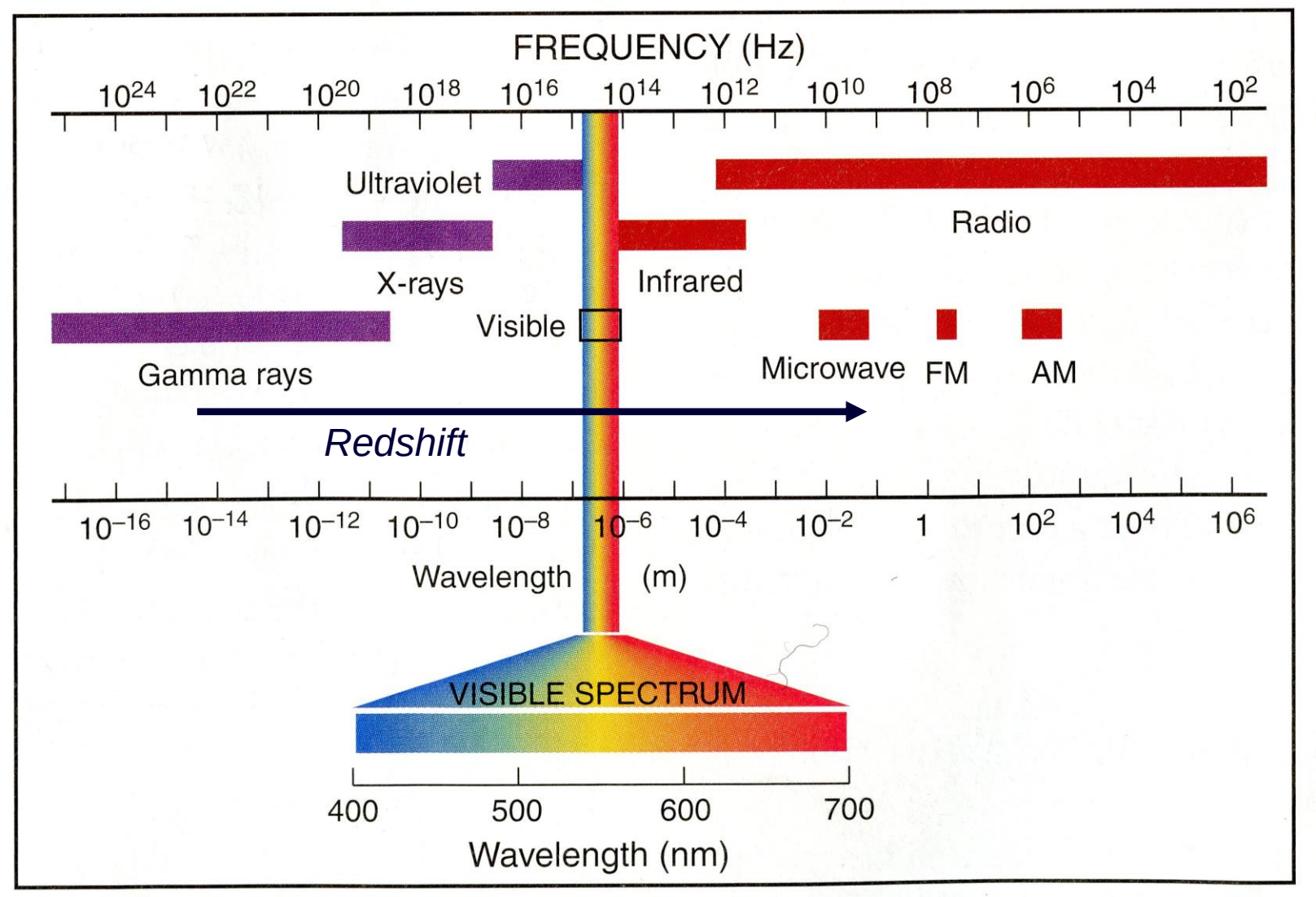
Cosmic microwave background: $t \sim 380$ kyr

The closest moment to the Big Bang that we can observe using photons is the time of photon decoupling when the Universe became transparent



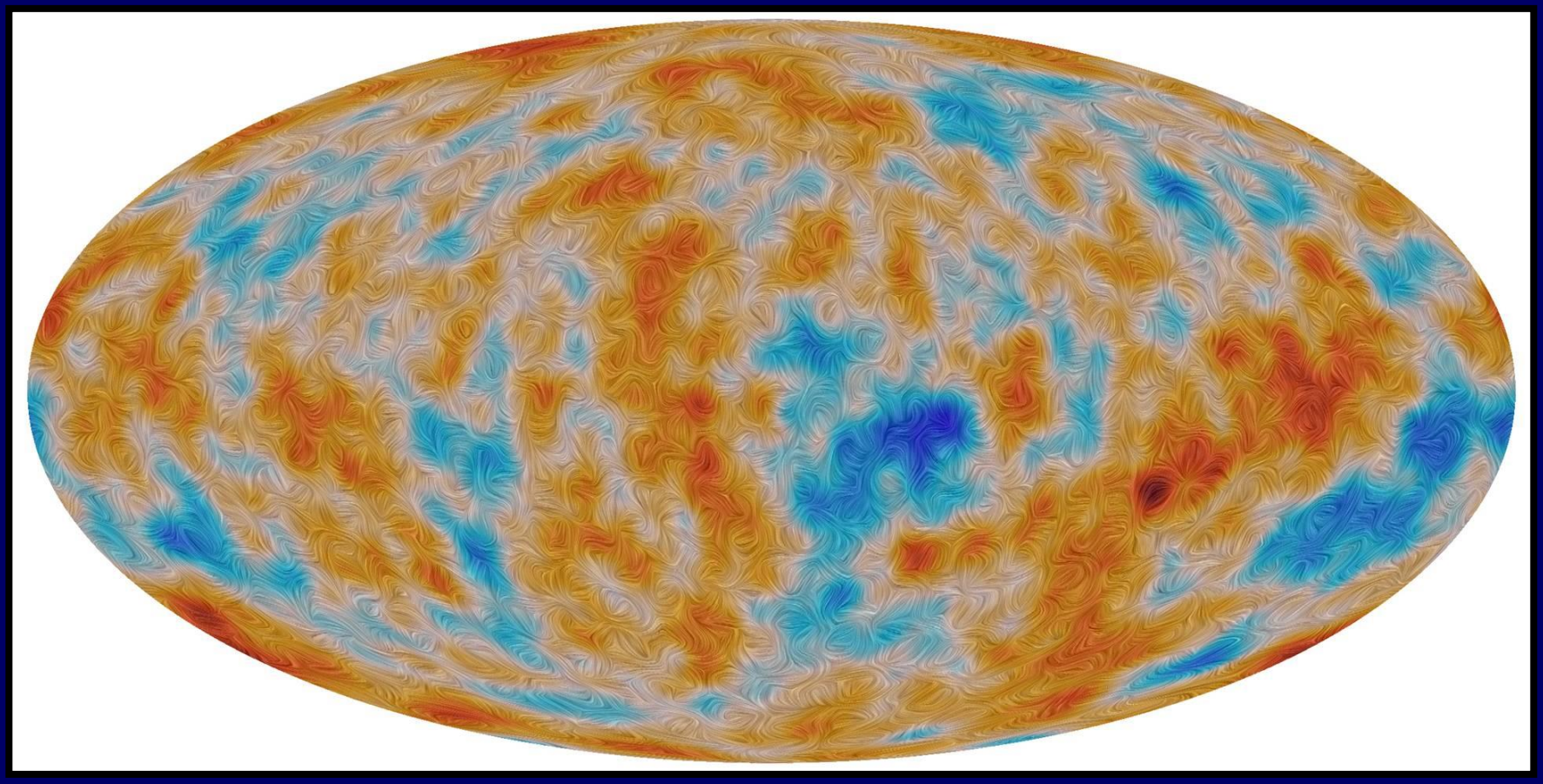
A map of the cosmic microwave background colour coded to show temperature variations across the Universe ~ 380 kyr after the Big Bang as the Universe became transparent.

Cosmic microwave background: $t \sim 380$ kyr



Cosmic microwave background: $t \sim 380$ kyr

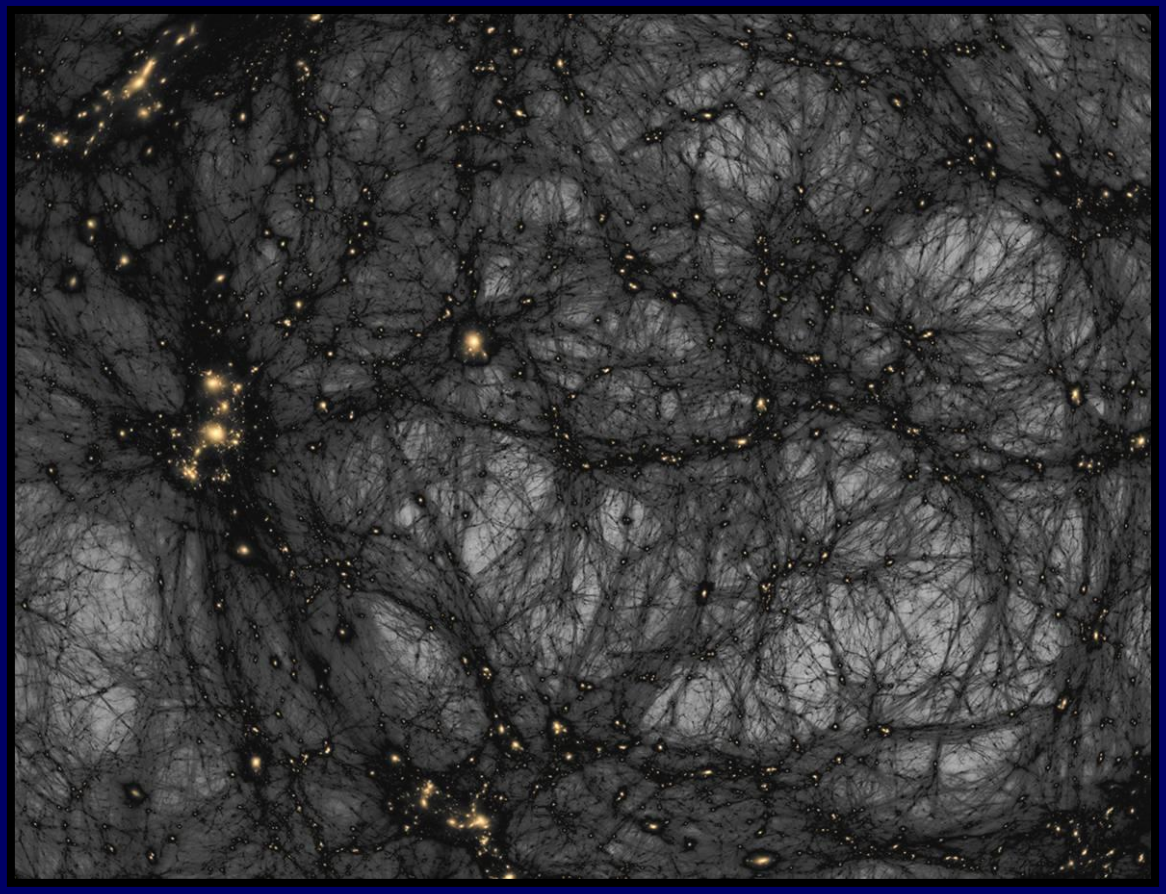
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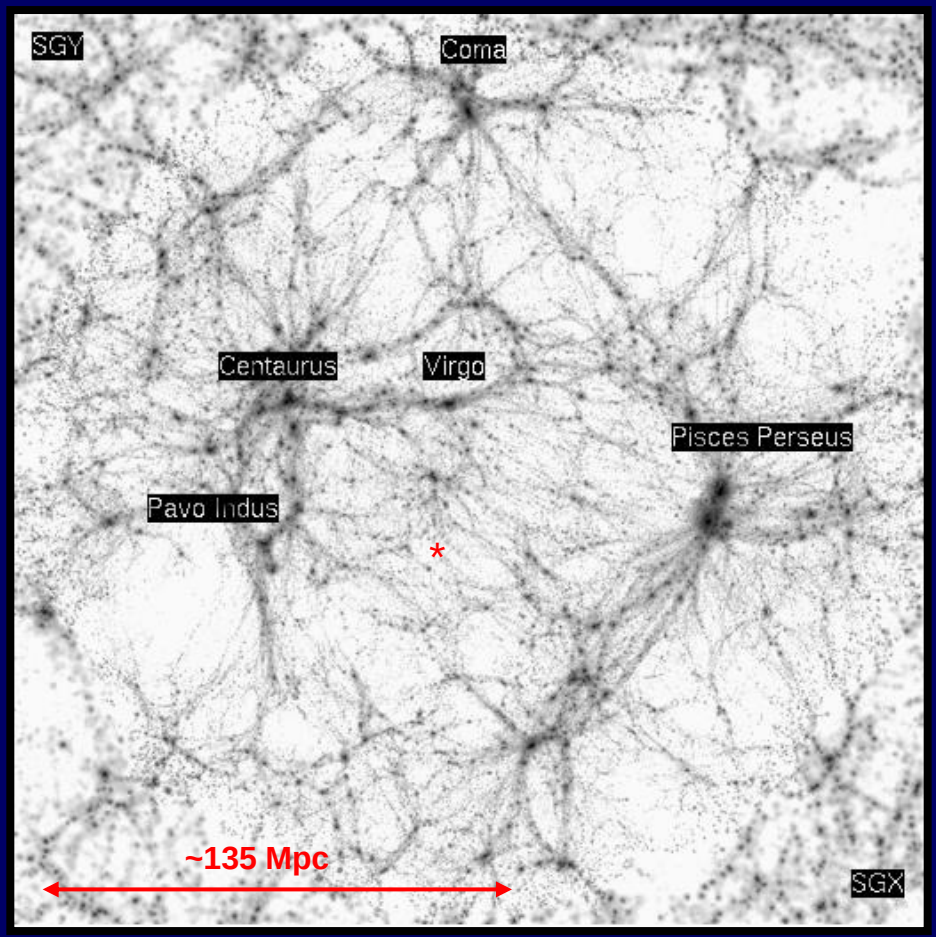
Cosmic Dark Ages: 380 kyr < *t* < 150 myr

- All the atoms and light in the universe make up <5% of the content.
- The rest is dark matter and dark energy, which are invisible but dominate the structure and evolution of the universe.
- Dark matter makes up most of the mass of galaxies and galaxy clusters, and is responsible for the way galaxies are organized on grand scales.
- Dark energy is driving the accelerated expansion of the universe.

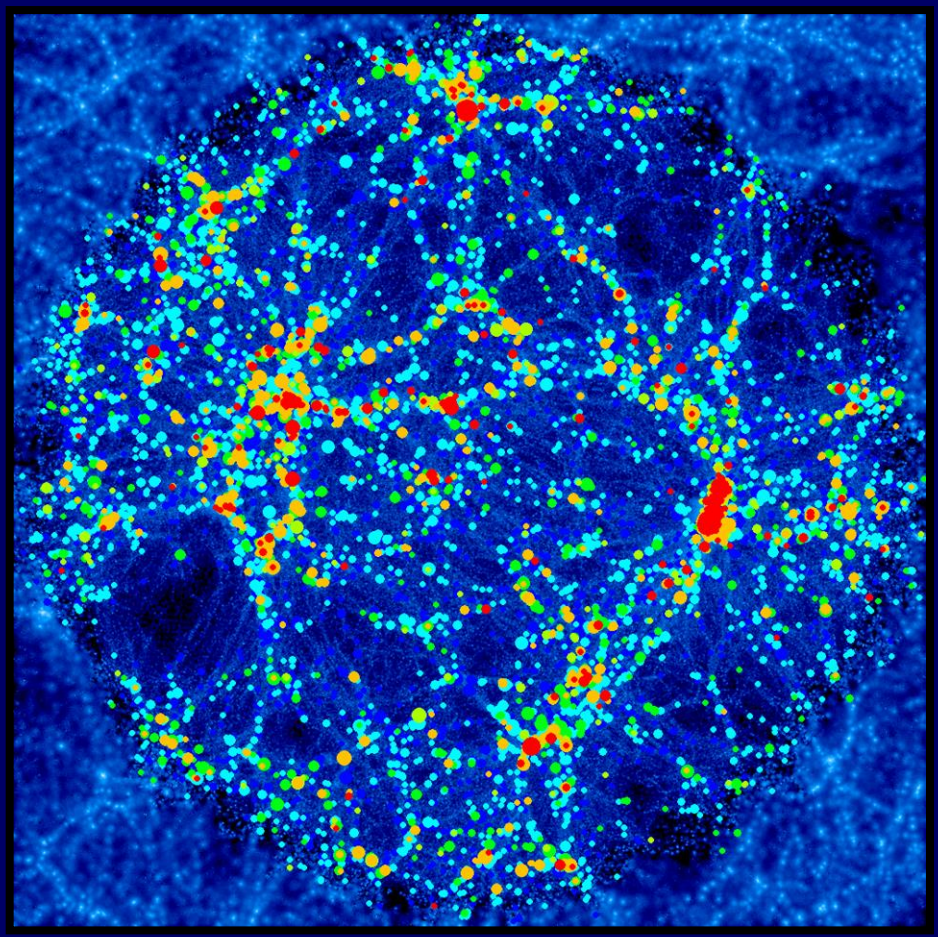


A simulation of the distribution of dark matter (dark areas) across a region of Universe

Effect of dark matter on galaxy distribution



Simulation of dark matter distribution across the nearby region of the Universe (red asterisk Milky Way galaxy)



Same area, with the distribution of galaxies observed today superimposed

Within the Λ-CDM model perspective there is insufficient normal matter to overcome the expansion of the Universe. Dark matter is essential for explaining the formation of structure in the Universe.

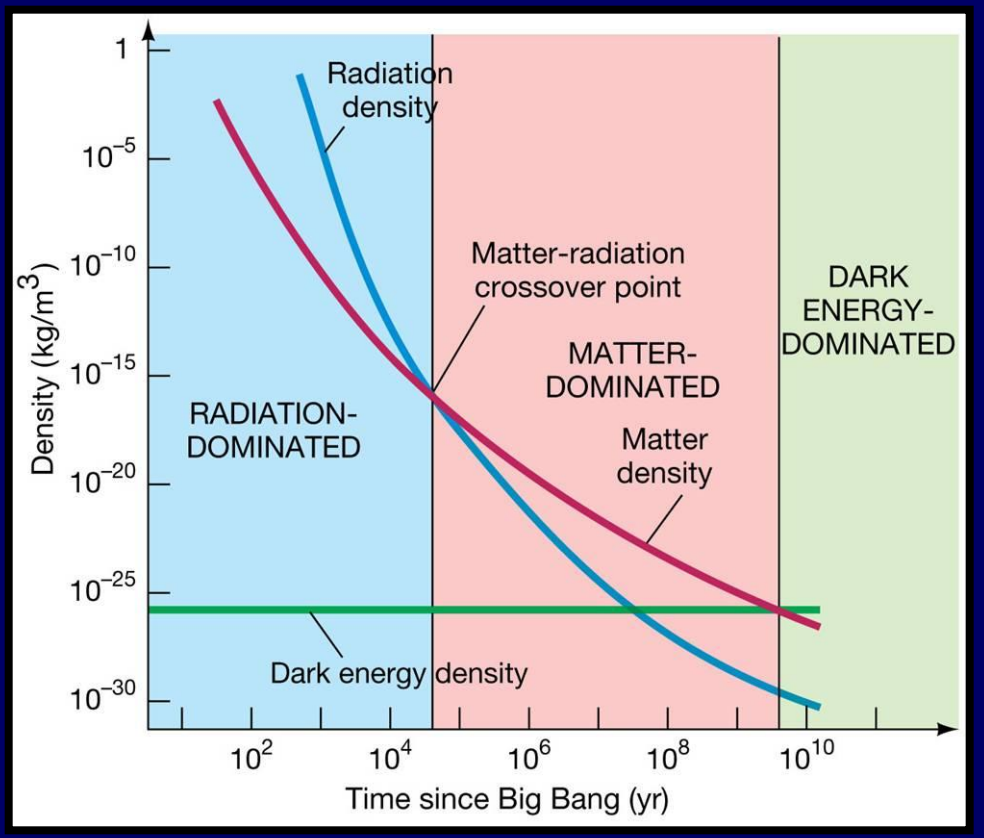
The future of expansion

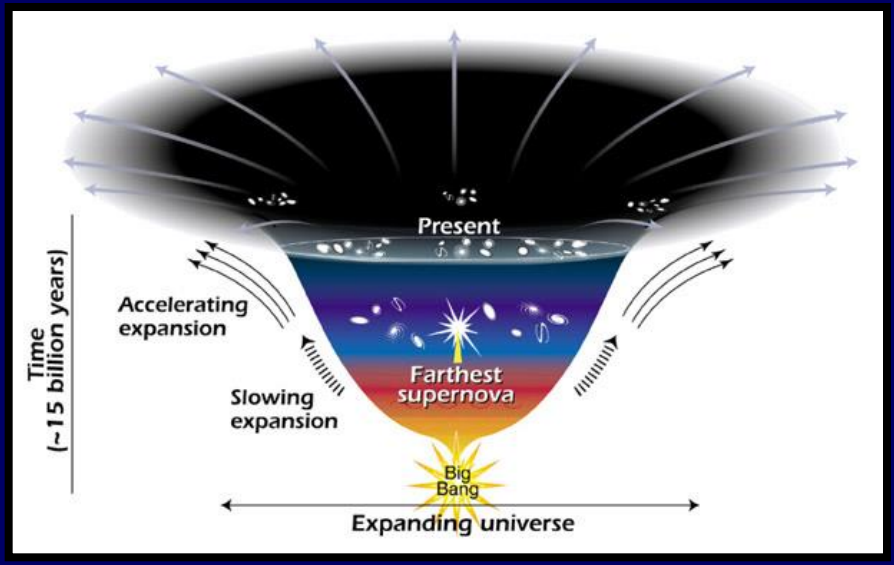
the Λ-CDM model

The currently observable Universe is estimated to comprise:

68.5% dark energy
26.6% dark matter
4.9% normal matter

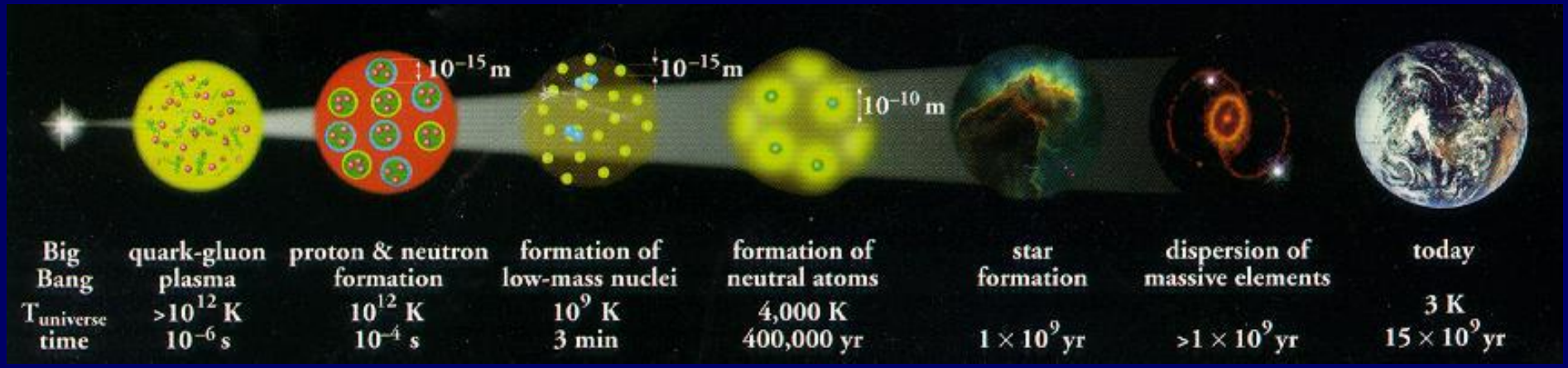
with only trace amounts of radiation





Dark energy has become increasingly dominant resulting in an accelerating expansion of the Universe beginning ~ 7500 years after the Big Bang

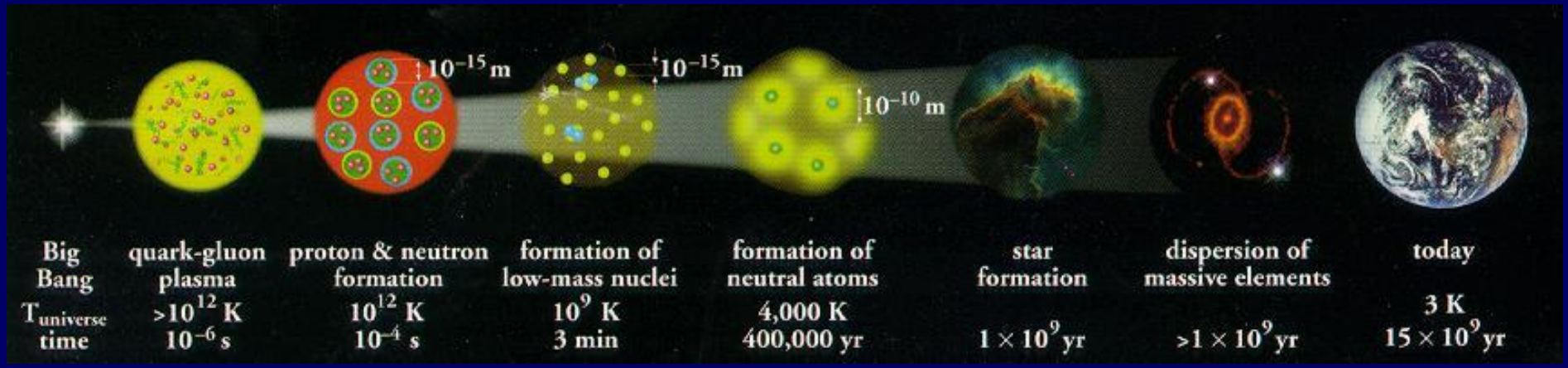
Intended learning outcomes revision



Now we have completed this class you should be able to:

- state the age of the Universe and explain how we know this
- conceptualise the Big Bang Theory and assess the evidence for the expanding Universe
- list the sequence of events that occurred during the first 150myr following the Big Bang
- describe where hydrogen, helium and lithium came from
- explain the significance of dark energy and dark matter within the Λ-CDM model and for the structure of the Universe

Independent study



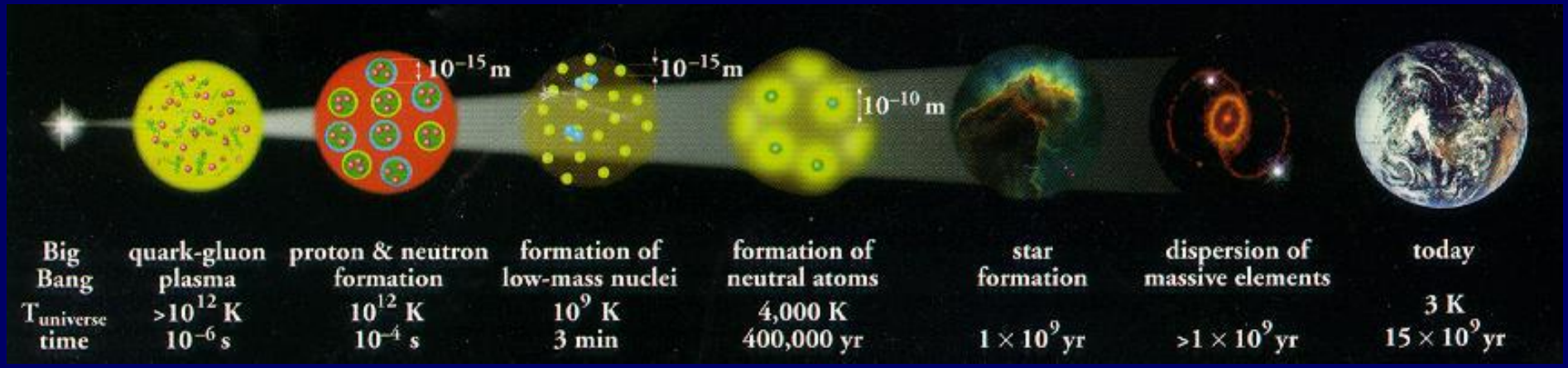
Suggested revision:

- Review the learning objectives for this week and write answers to the study guide questions on Blackboard
- Consolidate by reviewing the recorded lecture
- Optionally, read additional texts for more in-depth understanding (suggestions on Blackboard)

If you have not done so already

- Watch the film ‘Cosmography of the Local Universe’ (link on Blackboard)
- Read Chapter 1 of Marshak (pdf on Blackboard)

Next week……



The Heavy Construction: Formation of the Stars and Galaxies

13650 - 4567 Ma

- The Birth of the First Stars
- The Life-cycle of Stars and origin of elements
- The Formation and Evolution of the Milky Way Galaxy